EDA and Data Analysis Report

Supply Chain Optimization Project

Aditya Sethi

iitmd ba 2501026

1. Objective

The objective of this analysis is to understand order patterns, customer demand behavior and other key factors that affect inventory and forecasting demands. Report was made using the e-commerce dataset downloaded from kaggle. This exploratory analysis forms the foundation for forecasting and inventory optimization.

2. Dataset Description

The dataset was taken from Kaggle consists of multiple relational tables including orders, order items, products, customers, and sellers. These tables were merged using primary keys such as order id and product id to build a master analytical dataset.

```
[7]
✓ 0s
tables = {
    "orders": orders,
    "items": items,
    "products": products,
    "customers": customers,
    "sellers": sellers
}

for name, df in tables.items():
    print(name, df.shape)

orders (99441, 8)
items (112650, 7)
products (32951, 9)
customers (99441, 5)
sellers (3095, 4)
```

3. Data Integration

On the raw dataset, the following joins were performed using google collab platform:

- Orders joined with Order Items on order id
- Products joined on product id
- Customers joined on customer id
- Sellers joined on seller id

```
[8]
master = orders.merge(items, on="order_id", how="left")

master = master.merge(products, on="product_id", how="left")

master = master.merge(customers, on="customer_id", how="left")

master = master.merge(sellers, on="seller_id", how="left", suffixes=("_cust","_seller"))

master = master.merge(cat_trans, on="product_category_name", how="left")

print(master.shape)
master.head()
```

```
(113425, 30)
```

	order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	delivered	2017-10-02 10:56:33	2017-10-02 11:07:15	2017-10-04 19
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27	2018-07-26 14
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23	2018-08-08 13
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59	2017-11-22 13
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29	2018-02-14 19

4. Data Cleaning

Data cleaning was done after merging the tables.

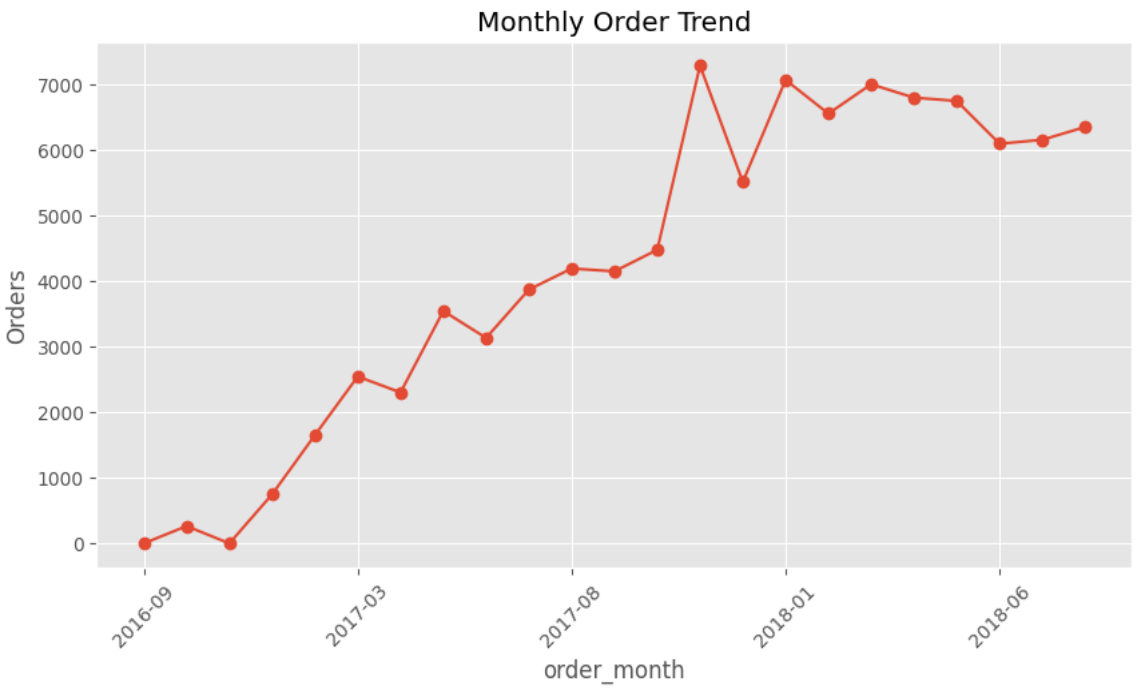
- Filtered only delivered orders
- Removed missing delivery dates
- Converted timestamps to date time
- Removed negative price and freight values
- Filled missing categories as “unknown”
- Created derived features: order value, month, weekday, hour

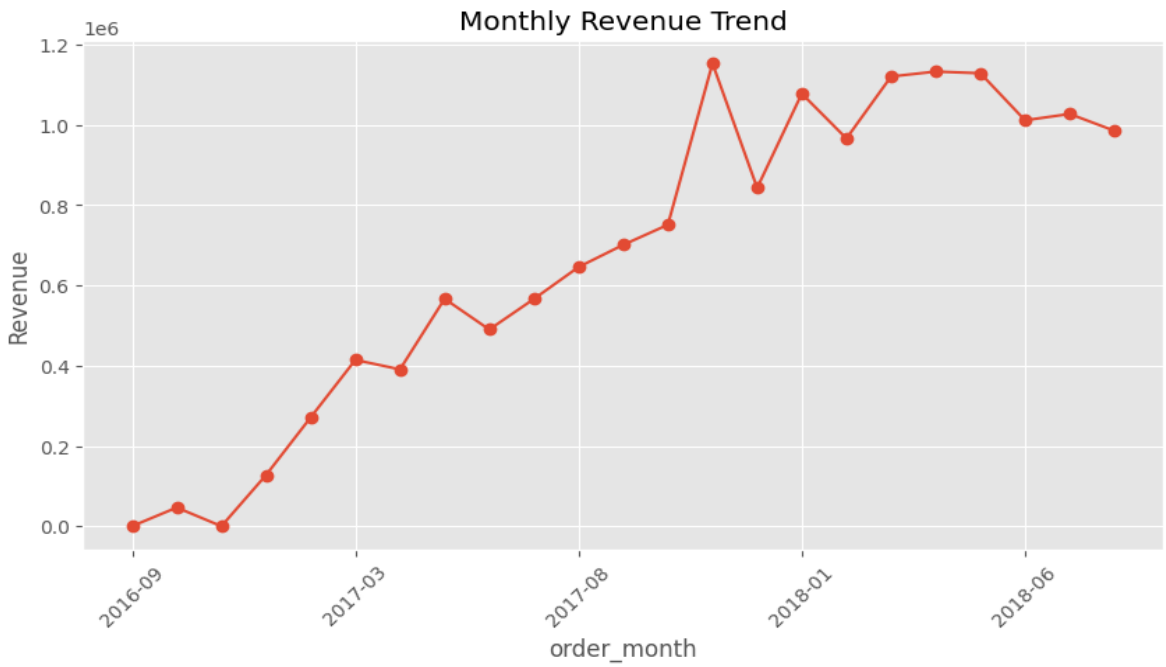
5. Key KPIs

Metric	Value
Total Orders	96470
Total Revenue	15418394.82
Average Order Value	159.82
Avg Items per Order	1.14

6. Order Trends

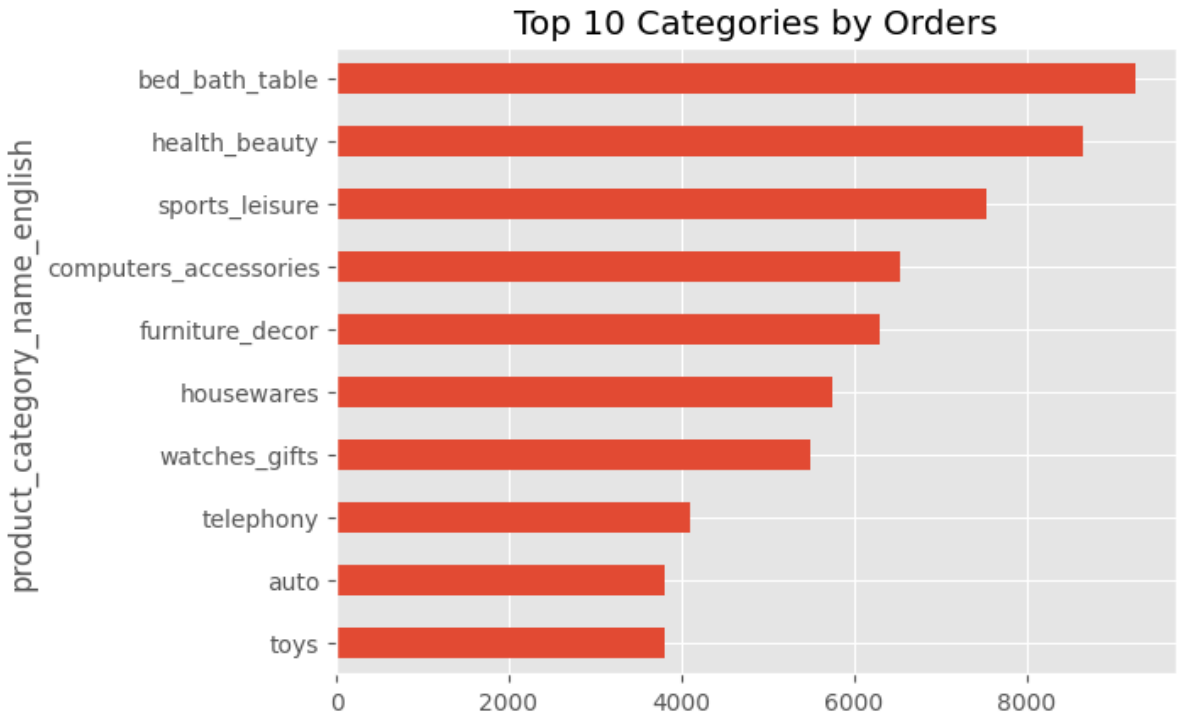
Monthly order and revenue trends show an overall growth pattern with visible seasonal peaks.

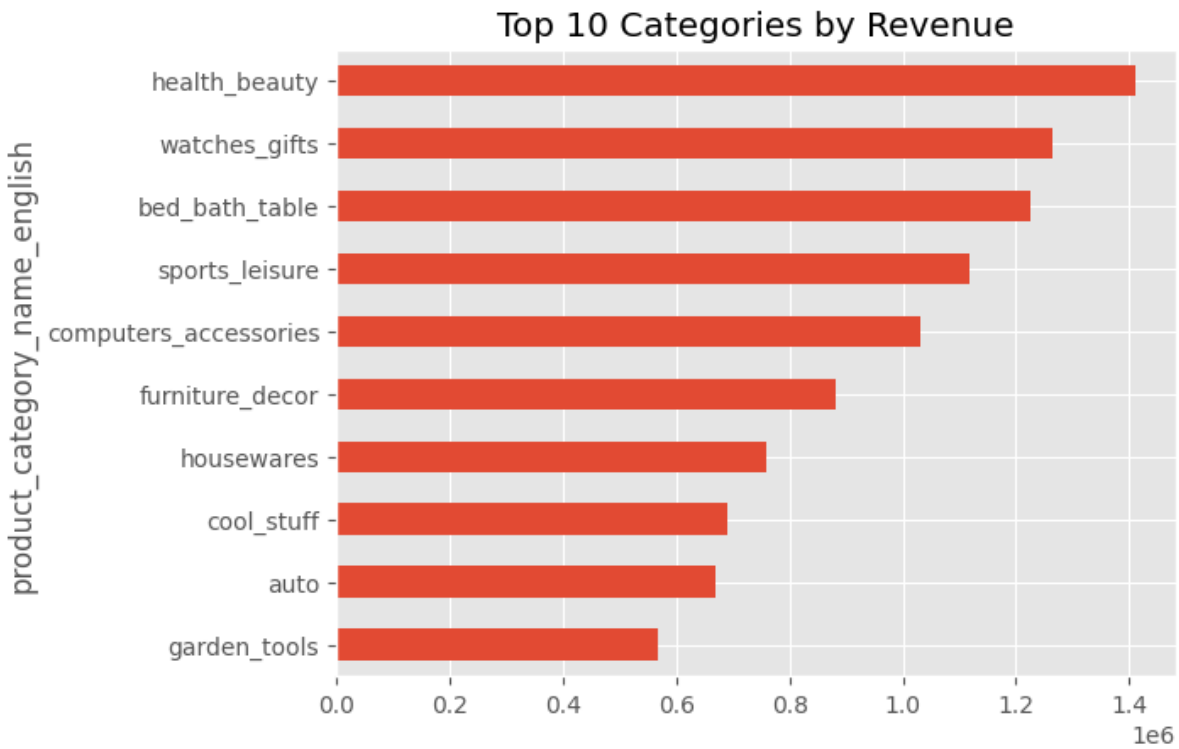




7. Category Analysis

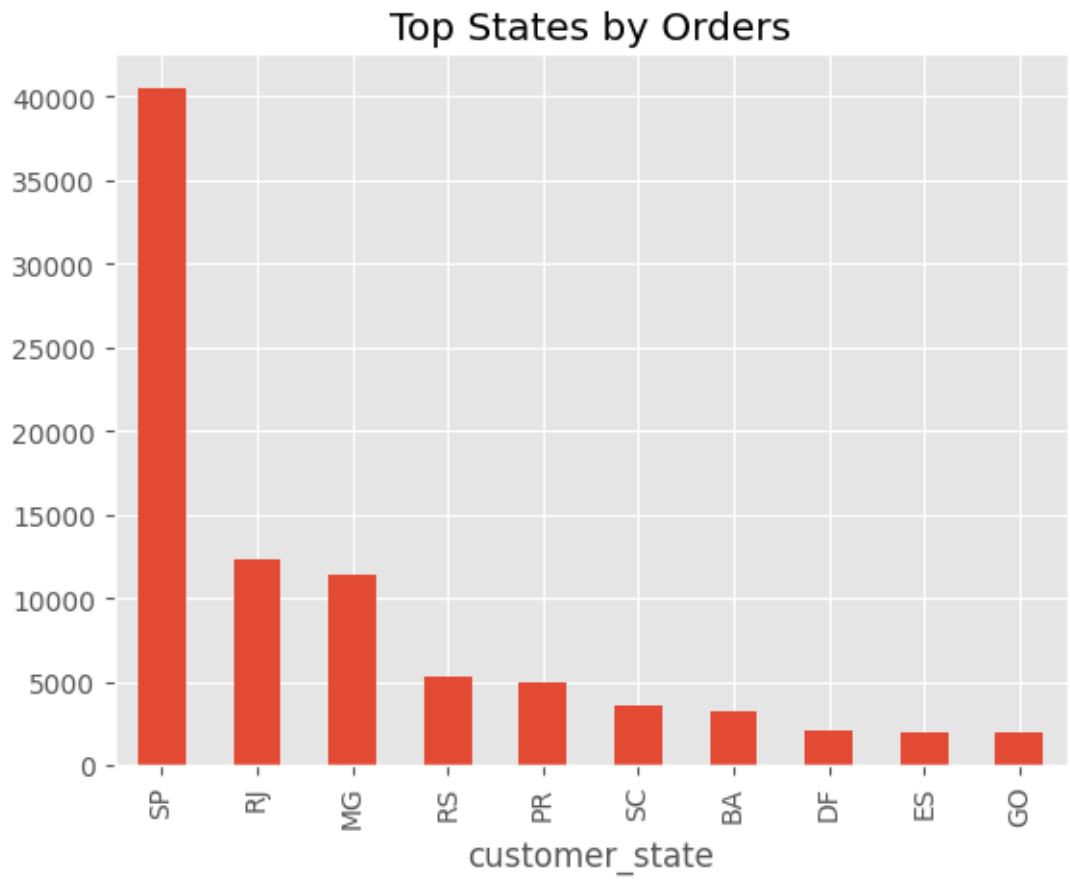
Top categories differ by volume and revenue, indicating varying price structures.





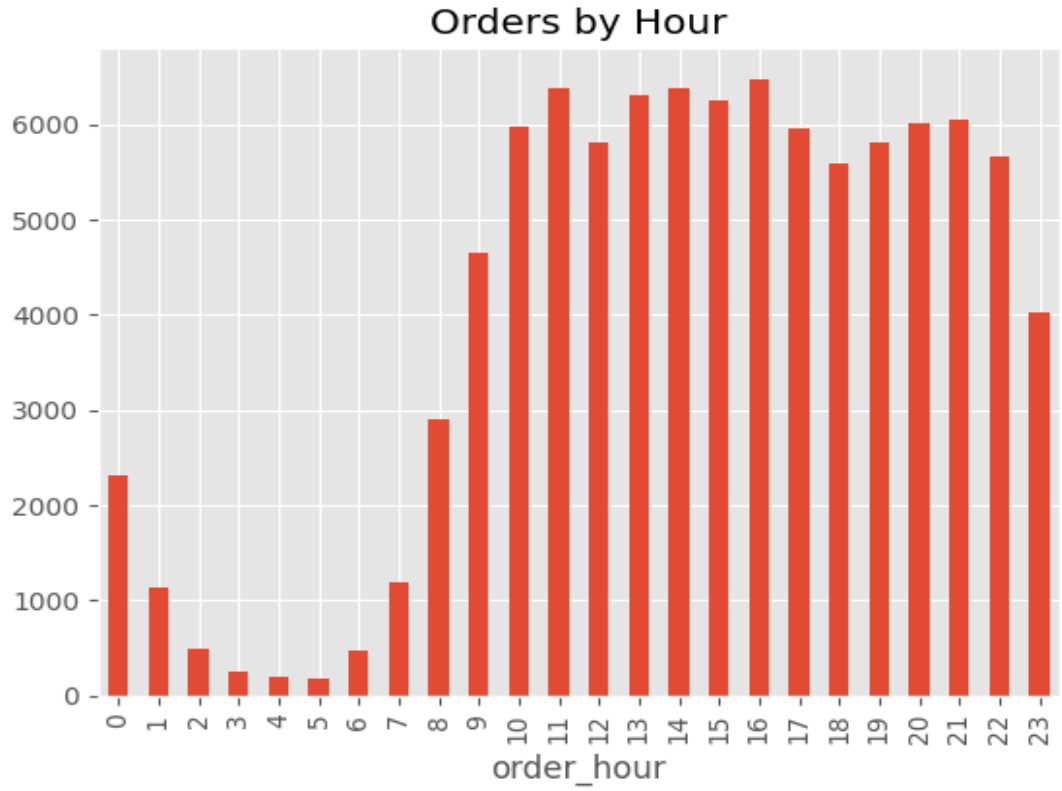
8. Geographic Analysis

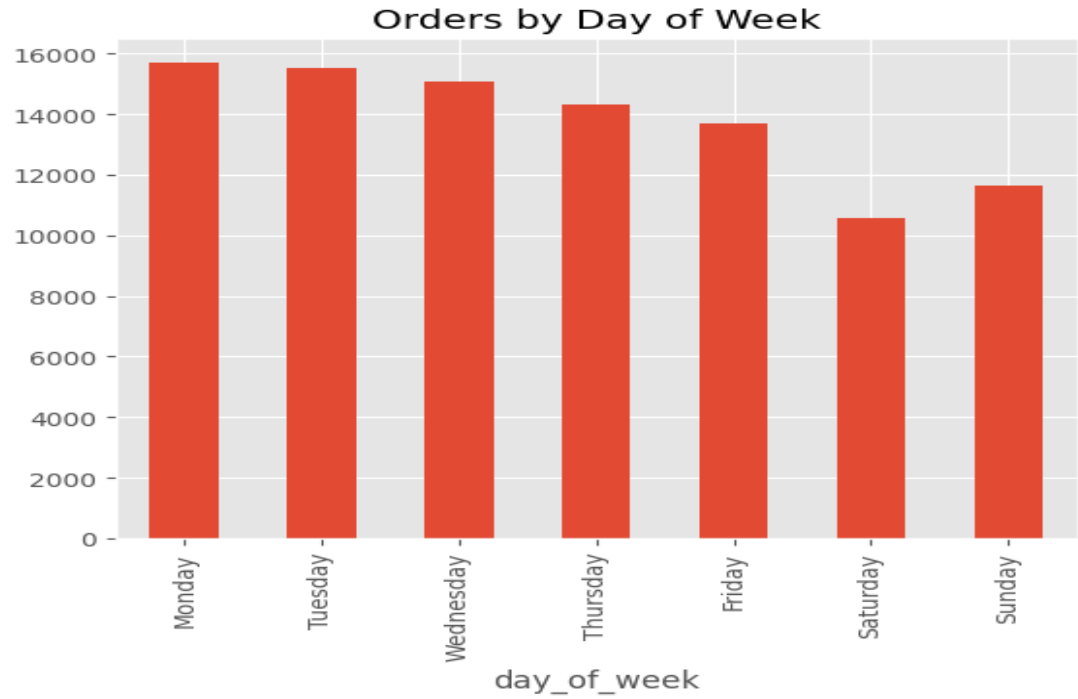
Demand is concentrated in major states and cross-state shipping is dominant.

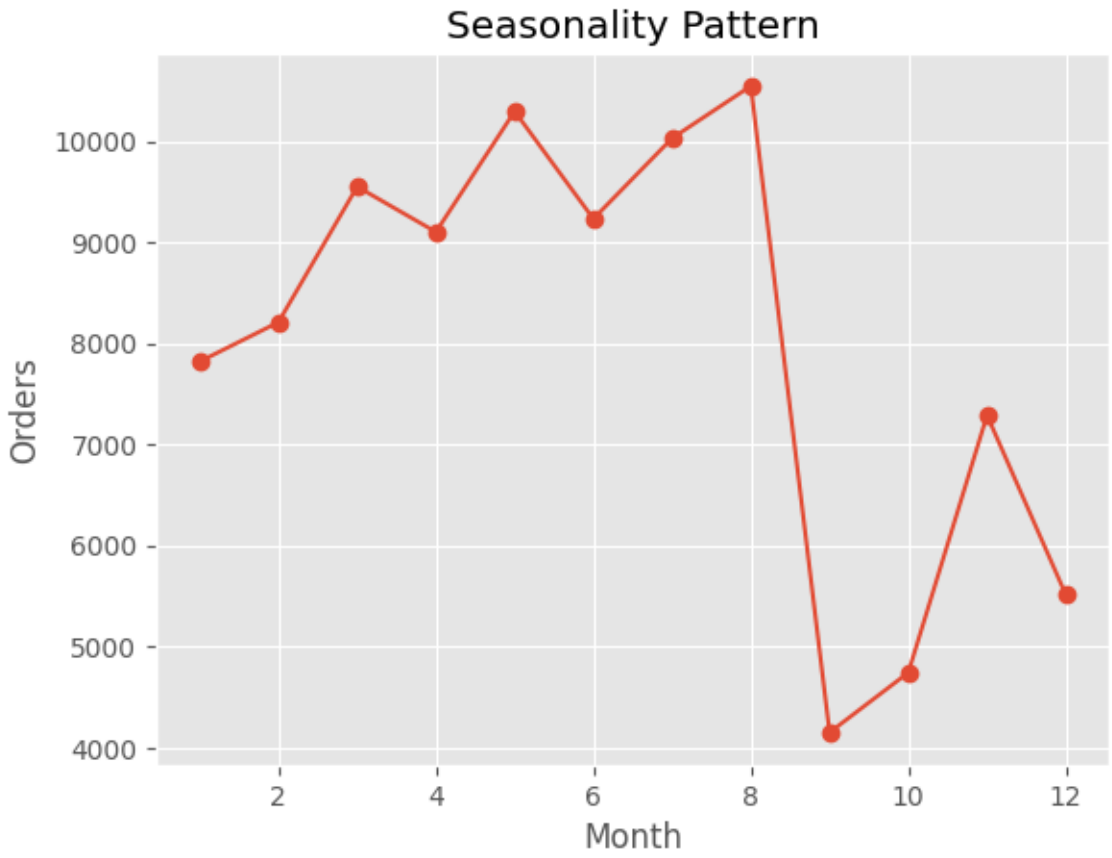


9. Time Pattern Analysis

Orders show weekday concentration and peak ordering hours.





10. Key Findings

- Upward demand trend observed
- Category concentration present
- Regional demand clusters identified
- Seasonal spikes detected